PROPRIETARY & CONFIDENTIAL — LWANDILE ENGINEERING PROJECTS & INCLUSIVE ENTREPRENEURIAL ACADEMY

SCAPUSH PRECISION STANDARD

SECTION 21 — DESIGN REGISTERS & GOVERNANCE

Sovereign Bio-Solar-Thermal Campus | Master Design Specification

Doc Ref: 21_governance.html | Status: MASTER ARCHITECTURE COMPLETE | Governance: Scapush Precision | Institutional Change Control

Auto-Scrolling Live Narration Engine

▶ Start Auto-Scroll Narration ■ Stop / Misa

1. Purpose & Governance Philosophy

LISTEN SECTION

This section establishes the institutional governance framework and change management structure for all engineering, operational, environmental, educational, and infrastructure decisions across the Sovereign Bio-Solar-Thermal Campus. The Design Registers provide a permanent, auditable, and traceable record of technical assumptions, baseline freezes, risk mitigations, revision logs, and commissioning milestones throughout the project lifecycle.

INSTITUTIONAL GOVERNANCE MANDATE

All major engineering parameters are documented, traceable, reviewable, auditable, and strictly version-controlled under the Scapush Precision Execution Standard. No design alteration, capacity scaling, or structural change shall occur without formal approval and entry into the appropriate master register.

2. Core Engineering & Design Registers

LISTEN SECTION

2.1 Design Assumptions Register

ID	Engineering Assumption Parameter	Governance Status
EA-001	Deep groundwater borehole yield available (1.5–2.0 m^3/h target)	Verified / Open
EA-002	20,000m^2 leasehold footprint secured for facility development	Verified / Open

2.2 Design Freeze Register

ID	Locked Master Design Parameter	Freeze Status
DF-001	20,000m^2 Master Site Boundary & Zoning Grid	FROZEN (Rev 1.0)
DF-002	Poultry Barn 1 Footprint (12m × 32m = 384m^2)	FROZEN (Rev 1.0)
DF-003	Swine Barn 2 Footprint (12m × 32m = 384m^2)	FROZEN (Rev 1.0)

ID	Engineering Assumption Parameter	Governance Status
EA-003	70m³ mesophilic digester capacity suitable for Phase 1 waste inputs	**Under Review**
EA-004	100 kWp rooftop + tracker solar expansion reserve adequate for Phase 2	**Under Review**

ID	Locked Master Design Parameter	Freeze Status
DF-004	500 Swine Weaner Capacity Scale (8 Pens)	**FROZEN (Rev 1.0)**
DF-005	5,720 Broiler Batch Capacity Scale	**FROZEN (Rev 1.0)**

2.3 Engineering Risk Register

ID	Identified Risk Vector	Engineering Mitigation Strategy
ER-001	Low Borehole Water Yield	Geophysical resistivity survey + 20,000L elevated storage (68h autonomy).
ER-002	Insufficient Digester Feedstock	Dual-source loading (Swine slurry + Poultry litter) + organic biomass supplement.
ER-003	Solar PV Underperformance	3-way sync: LFP BESS discharge + Biogas CHP baseload generation.
ER-004	Pathogen / Disease Outbreak	10-Layer biosecurity system (Shower-in/out, vehicle spray arch, pod isolation).

2.4 Missing Items Register (Pre-Lockout Review)

ID	Engineering Item Requiring Detailed Study	Current Review Status
MI-001	Detailed Lightning Protection Study (SANS 62305)	**Pending Phase 2**
MI-002	Dedicated Fire Water Tank Hydrant Ring Main Study	**Pending Phase 2**
MI-003	Gas Detection Sensor Spatial Layout Plan	**Pending Phase 2**
MI-004	Full Campus Mass & Energy Balance Calculation (Doc 32)	**Pending Phase 2**
MI-005	Full Campus Water Balance Calculation (Doc 31)	**Pending Phase 2**

2.5 Institutional Change Control Register

Change ID	Proposed Engineering Revision / Modification	Technical Impact & Approval Justification	Status
CR-001	Swine Capacity Scaling from 160 to 500 Weaners	Increased manure yield supporting continuous 70m³ digester loading and 15–25 kW CHP.	**APPROVED**
CR-002	Integration of Zone M Accessibility & EV Hub	Embedded 30 kW DC EV charger, wheelchair charging bays, and universal step-free access.	**APPROVED**

3. Operational, Environmental & Commissioning Registers

LISTEN SECTION

3.1 Asset Register Framework

- **Energy Assets:** Solar PV Arrays, LFP BESS Bank, Biogas CHP Engine, Microgrid Switchgear.
- **Water Assets:** Borehole-01, Submersible DC Pump, Treatment Plant, Elevated Tanks.
- **Production Assets:** Poultry Pods, Swine Pen Arrays, Nursery Hydronic Manifolds.
- **Digital Assets:** SCADA Servers, Industrial PLCs, Fiber Optic Network, Sensors.

3.2 Commissioning Register

- **Phase 1:** Mechanical Completion & Structural Inspection.
- **Phase 2:** Electrical & Microgrid Bus Energization.
- **Phase 3:** Water Treatment & Hydrostatic Pressure Testing.
- **Phase 4:** Biodigester Slurry Seeding & CHP Synchronization.
- **Phase 5:** SCADA Telemetry & Full System Integration.

3.3 Operational Readiness

- **Staff Certification:** Operators certified via Academy bootcamps.
- **SOP Approval:** Standard Operating Procedures signed for all assets.
- **Emergency Protocol:** Emergency response and evacuation plans active.
- **Biosecurity Active:** Shower-in/out and vehicle arches operational.

4. Master Design Governance Freeze Register (Sections 01–20)

LISTEN SECTION

The core structural and technical specifications across all twenty infrastructure sections are locked as approved baselines:

Ref Code	Locked Infrastructure Package	Governance Basis & Document Reference	Status
DG-001	Document Control & Framework	Scapush Precision Standard (Doc Ref: 01_framework.html)	**FROZEN**
DG-002	Executive Summary & Vision	Sovereign Campus Mandate (Doc Ref: 02_summary.html)	**FROZEN**
DG-003	Design Basis Memorandum	Master System Parameters (Doc Ref: 03_design_basis.html)	**FROZEN**
DG-004	Site Strategy & Selection	20,000m² Leasehold Site (Doc Ref: 04_site_strategy.html)	**FROZEN**
DG-005	Master Site Layout & Zoning	Zones A through M Layout (Doc Ref: 05_master_layout.html)	**FROZEN**
DG-006	Poultry Cluster Package	Barn 1: 5,720 Broilers (Doc Ref: 06_poultry_package.html)	**FROZEN**
DG-007	Swine Cluster Package	Barn 2: 500 Weaners (Doc Ref: 07_swine_package.html)	**FROZEN**
DG-008	Climate Controlled Nursery	Zone F Dual Hydronic Loop (Doc Ref: 08_nursery_package.html)	**FROZEN**
DG-009	Biosecurity Architecture	10-Layer Biosecurity Grid (Doc Ref: 09_biosecurity.html)	**FROZEN**
DG-010	Water Sovereignty Architecture	Borehole-01 & 20,000L Storage (Doc Ref: 10_water_architecture.html)	**FROZEN**

Ref Code	Locked Infrastructure Package	Governance Basis & Document Reference	Status
DG-011	Biodigester Architecture	70m³ Anaerobic Digester (Doc Ref: 11_biodigester.html)	**FROZEN**
DG-012	CHP Thermal Architecture	15–25 kW Biogas Engine (Doc Ref: 12_chp_thermal.html)	**FROZEN**
DG-013	Solar PV Architecture	30–60 kWp Trackers & Roof (Doc Ref: 13_solar_pv.html)	**FROZEN**
DG-014	BESS Storage Architecture	120–200 kWh LFP Storage (Doc Ref: 14_bess.html)	**FROZEN**
DG-015	Microgrid Architecture	Off-Grid Power Dispatch (Doc Ref: 15_microgrid.html)	**FROZEN**
DG-016	SCADA & Digital Operations	Central Digital Platform (Doc Ref: 16_scada.html)	**FROZEN**
DG-017	Accessibility & Inclusion	Zone M Hub & Universal Access (Doc Ref: 17_accessibility.html)	**FROZEN**
DG-018	Entrepreneurship Academy	Human Capital Engine (Doc Ref: 18_academy.html)	**FROZEN**
DG-019	Environmental Stewardship	Regenerative Circular Loop (Doc Ref: 19_environmental.html)	**FROZEN**
DG-020	Emergency Response	Life Safety & Failsafes (Doc Ref: 20_emergency.html)	**FROZEN**

5. Next Engineering Phase & Detailed Quantification Sequence

LISTEN SECTION

With Sections 01 through 21 formally approved and frozen, the master design moves into the detailed engineering, mass/energy balancing, equipment sizing, and P&ID development sequence:

Detailed Engineering Phase Deliverables Sequence

Document Reference	Detailed Engineering Phase Deliverable	Technical Target Scope
31_water_balance.html	Master Campus Water Balance Calculation	Detailed volumetric mass balance across borehole, tanks, pods, digester & losses.
32_energy_balance.html	Master Campus Mass & Energy Balance	Thermal & electrical kW/kWh balancing across Solar, BESS, CHP, loads & heat loops.
33_equipment_sizing_register.html	Master Equipment Sizing Register	Mechanical, electrical & hydraulic sizing specifications for all pumps, motors & tanks.
34_instrumentation_register.html	Master Instrumentation Register	Comprehensive tag list for all field sensors, transmitters, flow meters & valves.
35_io_schedule.html	SCADA PLC Input / Output Schedule	Digital & Analog I/O point allocation for all local PLCs and central SCADA nodes.
36_p_and_id_basis.html	Piping & Instrumentation Diagram Basis	P&ID schematics for water reticulation, gas piping, hydronic loops & slurry lines.

6. Final Master Architecture Declaration

LISTEN SECTION

FINAL DECLARATION & INSTITUTIONAL SEAL

The Sovereign Bio-Solar-Thermal Campus is hereby established as a Food-Energy-Water-Accessibility Nexus Demonstration Platform, integrating regenerative agriculture, renewable energy, resource recovery, digital intelligence, inclusive entrepreneurship, and environmental stewardship into a single sovereign ecosystem.

7. Approval & Authorization

LISTEN SECTION

DOCUMENT STATUS

STATUS: MASTER DESIGN SPECIFICATION CORE ARCHITECTURE COMPLETE (REV 1.0) — APPROVED FOR DETAILED ENGINEERING PHASE

PROJECT STEWARD & TECHNICAL LEAD



Select File Lock Signature

Sipho Wiseman Khuzwayo
Inclusive Legacy Architect / Principal Lead
Lwandile Engineering Projects (Pty) Ltd
Status: Signature Image Loaded

GOVERNANCE & ACADEMIC LEAD

Click or drag & drop signature file
(JPG / PNG / SVG)

Select File Lock Signature

Dr. Palesa Charlotte Felix-Faure
Founder & Academic Lead
Inclusive Entrepreneurial Academy
Status: Awaiting Signature File

Consortium Leadership

Sipho Wiseman Khuzwayo — Lead System Architect (Lwandile Engineering)
Dr. Palesa Charlotte Felix-Faure — Academic & Governance Lead (Inclusive Entrepreneurial Academy)

Execution Entities & Standard

Lwandile Engineering Projects (Pty) Ltd
Inclusive Entrepreneurial Academy
Scapush Precision Execution Standard

Operational Footprint

Sovereign Bio-Solar-Thermal Campus, Mpumalanga, South Africa

Primary Contact & Communication

siphokhuzwayo12@gmail.com
LinkedIn Profile